

Speech-Based Emotion, Age and Gender Detection with Speech-to-Text

Roopa Ramthapure

Dept of CSE

Vardhaman College of Engineering

Hyderabad, India.

rooparamthapure@gmail.com

Dhudukala Thapaswini

Dept of CSE

Vardhaman College of Engineering

Hyderabad, India.

dhudukalathapaswini@gmail.com

Vemula Sai Prasaad

Dept of CSE

Vardhaman College of Engineering

Hyderabad, India.

saiprasaad9100@gmail.com

Dr. U Sesadri

Dept of CSE

Vardhaman College of Engineering

Hyderabad, India.

seshadri1.u@gmail.com

Dr.K.Madhavi

Dept of Mathematics

Malla Reddy(MR) Deemed to be University

Hyderabad, India.

madhavimathematics@gmail.com

Abstract—Speech signals are a rich source of information on the emotional condition, age, gender and content being spoken by the speaker and, thus, automatic speech analysis is becoming an essential field of research in applications like human computer interaction, smart voice system, and medical monitoring systems. Nevertheless, it is not easy to recognize several speaker features based on audio cues because of voice differences and recording environments. In this paper, the proposal is a hybrid machine learning and deep learning system that will classify emotions, age, and gender based on speech with the input of speech-to-text conversion. The system makes use of the RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and song) in training and evaluation. A variety of acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), delta and delta-delta coefficients, chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, spectral centroid, spectral rolloff, and pitch features are extracted with the help of Librosa library. In the case of emotion recognition, a deep learning network based on Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) networks to track both spatial and temporal speech signal features. In the case of gender and age classification, the XGBoost (Extreme Gradient Boosting) classifiers are applied to be able to learn well-structured features representations. Also, there is a speech-to-text application built on Google Speech Recognition API that transfers audio speech to text. The achieved results of the experiment prove that the hybrid framework proposed by the authors delivers promising performance in predicting emotions, gender, and age, which allows concluding about the efficiency of integrating machine learning and deep learning methods into the single speech analysis systems.

Keywords: Speech Emotion Recognition, Age prediction, Gender detection, machine learning, deep learning, CNN-BiLSTM, XGBoost, and Speech-to-Text.

I. INTRODUCTION

One of the most natural methods of communication in human beings is that of speech and there is much more information on speech as compared to verbal communication. Voice of an individual can help define his/her emotional status, age, and sex among other important characteristics of understanding human behavior and the improvement of human-machines communication. The automatic

analysis of speech is a useful research topic due to the rapid progress of voice technologies and artificial intelligence. The systems capable of understanding the feelings and demographic information including virtual assistants, calls center analysis, medical surveillance, and human-computer interaction are useful in emotion and demographic speech recognition.

However, speech signals are difficult to identify a number of characteristics of speakers. The speech might vary significantly in relation to the accent of the speaker, his style of talking, the circumstances under which the speech is recorded and the surrounding noise [1]. The fact that such diversities imply that the traditional systems will not be able to decode speech data with any accuracy. The speech analysis systems can therefore be improved by integrating the machine learning and deep learning technologies.

This work develops a system that takes control of analyzing speech automatically and identifying different characteristics of a speaker, through an audio signal. The system is geared towards the perception of the feeling, gender, age group and also the translation of speech to the text. To achieve this objective, the specified system consists of a great number of essential components.

- 1) To obtain vital speech characteristics like MFCC, chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, and pitch out of audio signals.
- 2) To create a deep learning model based on CNN and BiLSTM to accurately detect emotions through speech cues.
- 3) To use machine learning based on the XGBoost to predict the gender and age group of the speaker.
- 4) To decode spoken audio and make it a written data with a speech-to-text module to enhance understanding of the speech content.
- 5) To develop a system of hybrid learning integrating the concept of deep learning and machine learning to enhance the overall performance of the system as well as prediction accuracy.

The paper is organized in the following way: Section II is the literature survey connected with the speech emotion recognition and speech analysis systems. Section III outlines the methodology that is proposed to be employed in terms of emotion recognition through speech, gender recognition, age recognition, and text-to-speech conversion. Section IV contains the results of the experiments and discussion, and the conclusion and the future scope are addressed in the Section V.

The given system is tested on the RAVDESS dataset, which also consists of the emotional speech of various speakers. The system offers a hybrid architecture of speech analysis by integrating deep learning methods of emotion recognition with machine learning algorithms of demographic prediction. The experiment findings demonstrate that the suggested strategy can successfully regulate emotion, gender, and age based on speech extensions as well as offer speech-to-text transformation.

Moreover, the designed system exhibits the possibilities of combining several speech analysis tasks into one framework. More intelligent and context-aware voice-based applications can be supported by such systems [2]. The technique may be further advanced in the future by adding more properties to the speakers and better speech recognition method to more complex real world applications.

II. LITERATURE SURVEY

Speech emotion recognition is a concept that has had numerous researches over the past few years as researchers attempt to make machines understand human emotions using voice patterns [3]. Earlier research in this field mainly focused on conventional machine learning schemes using hand designed audio images. The typical methods that could be applied into classifying the emotions of the extracted speech features of Mel-Frequency Cepstral Coefficients (MFCC), the pitch, and energy were Support Vector Machines (SVM), Decision Trees, and Random Forest [4]. These methods were quite precise but often not able to discover the multifaceted time-changing fluctuations of speech signals.

With the advent of deep learning, the (speech) analysis through the form of the neural network began to be applied. The popularity of Convolutional Neural Networks (CNN) was of the possibility to have the network learn informative pattern of features of the spectrogram-based representation of audio on its own [6]. The emotion of speech recognition has been successfully deployed with the use of CNN models with training of spatial patterns of audio signal. However, despite the fact that CNN models are highly useful as regards to capturing local pattern of features they are not as useful as modeling long-term time dependence of speech sequences.

To address this downside, researchers started combining CNN and Recurrent Neural Networks (RNN), namely, with Long Short-Term Memory (LSTM) networks [7]. The LSTM models are meant to acquire sequential data and long-term reliance of a time-dependent data e.g. speech signals. CNN and LSTM combination can help the system get time-dependent and space-dependent speech features [8]. Other researchers have proved that hybrid models such as CNN-LSTM or CNN-BiLSTM are significantly more effective in the emotion recognition as compared to the traditional methods.

In addition to the emotion identification, speech is also used to infer demographics e.g. gender or age. Machine learning algorithms that have been highly used to accomplish these tasks include XG-Boost, Random Forest and Gradient Boosting since they can easily handle structured features representation. These models utilize the acoustic properties of speech signals in order to identify demographic properties. Gradient boosting algorithms are often able to work with relatively small datasets and structured feature vectors in comparison with deep learning models.

Busso et al. launched the IEMOCAP dataset that has proven to be one of the popular emotion recognition datasets in speech. The dataset consists of multimodal recording such as audio and video data that assists the researchers to study emotional expressions efficiently. Likewise, Livingstone and Russo have created the RAVDESS dataset on which emotional speech samples of high quality are presented to train and test emotion recognition systems.

The most recent researches have also looked into the manner in which speech analysis can be incorporated with other functionalities such as speech to text conversion, as well as multi task learning systems. These systems are meant to perform a variety of speech related functions with a single architecture which will

TABLE I
SUMMARY OF EXISTING STUDIES ON TRAFFIC PREDICTION

Ref	Summary of Findings	Methodology Used	Limitations	Accuracy/Performance
[1]	Demonstrated that the emotion of speech can be classified with the help of custom-made features with the help of traditional machine learning models.	Support Vector Machine (SVM) of pitch and MFCC.	There is the weakness to describe temporal constraints in speech signals.	About 70% in establishing emotion.
[2]	Integrated CNN characteristic and LSTM temporal sequence learning with improved outcome in emotion classification.	Hybrid CNN-LSTM based on spectrogram.	More complicated calculation and preparation.	82% classification of emotion.
[3]	Demonstrated that ensemble machine learning models can be useful in gender and emotion classification.	MFCC and energy feature MFCC and energy feature random forest classifier.	Complex audio patterns and big sets of features reduce the performance.	75% classification accuracy.
[4]	The models of deep learning are able to enhance speech emotion recognition because they learn more complicated patterns using audio data.	MFCC and spectral features Deep Neural Network (DNN).	Demands bulk data to be more generalized and perform.	Approach to 85% emotion recognition accuracy.
[5]	Shown that CNN models can be trained to learn to recognize spatial patterns on the basis of audio information on their own and increase the accuracy of emotion recognition.	CNN using spectrogram and MFCC representations.	CNN is not effective in attempting to model long term change in time series of speech in isolation.	Probability of recognition of emotion is approx 78%.

be more efficient and easy to use [9]. The intelligent systems can understand the content and context of spoken communication better with application of speech recognition along with emotion and demographic analysis. Such integrations are particularly beneficial in their daily use like smart assistants, call center monitoring, and health support systems. Therefore, the invention of hybrid models, consisting of machine learning and deep learning algorithms. Proposed work

employs this weakness by creating a more unified system of both the emotion recognition and gender detection and age prediction, and speech-to-text conversion to offer a more thorough speech analysis platform.

III. THEORETICAL BACKGROUND

Speech processing is a field of academic study that entails examination of speech cues in human beings and how to interpret them using computation techniques [10]. A speech signal contains numerous types of information of the linguistic content and features of the speaker including emotion, age and gender. This usually involves deriving meaningful acoustic cues of sound signals which depict different aspects of speech.

It is quite significant in feature extraction in speech analysis system. The acoustic features tend to be the representation of speech signals. Mel-frequency Cepstral Coefficients (MFCC) are the most popular ones because they are the speech processing features best able to explain the perceptual characteristics of a human hearing [11]. The other interesting features are chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, spectral centroid, spectral rolloff and features which are associated with pitch [12]. These attributes help in the apprehension of different properties of speech such as frequency distribution, energy variations and tonal attributes, which are discernible in determining properties of speakers.

Machine learning algorithms are also extensively applicable to speech analysis because they are applied in classification. XGBoost, Random Forest and Support Vector Machines algorithms are capable of learning patterns with organized feature vectors that the speech signals extract [13]. XGBoost is a gradient boosting algorithm that is applied to build successive decision trees to maximize the performance of prediction.

The deep learning models have now been popularized in speech analysis because they automatically acquire an elaborate representation of raw data or processed data. Convolutional Neural Networks (CNN) are generally used when extracting spatial patterns, and Long Short-Term Memory (LSTM) networks are developed in line with temporal interconnections in sequence data, e.g., speech signals. BiLSTM operates on both forward and backward directions of speech sequences, in such a manner the model is able to have a more insightful understanding of the context of the speech signal [14]. Jointly applying CNN and BiLSTM, one can acquire the spatial and temporal characteristics of the speech into the system, which would make the emotions recognition systems more effective.

IV. PROPOSED METHODOLOGY

The system suggested is aimed to identify various speaker characteristics like emotion, gender, and age and translate speech to text by the help of analyzing speech signals [15]. The audio information employed in this research is gathered in RAVDESS (Ryerson Audio-Visual database of emotional speech and song dataset), which consists of the records of emotional speech by various speakers. The audio files are loaded in the first stage and significant acoustic features are extracted with the help of the Librosa library [16]. Some of the extraction features are Mel-Frequency Cepstral Coefficients (MFCC), delta and delta-delta coefficients, chroma features, mel-spectrogram, spectral contrast, zero-crossing rate, spectral centroid, spectral rolloff, and pitch-related features. .

Both machine learning and deep learning models are used in the second step in classifications. Emotion recognition is performed with the help of a CNN-BiLSTM deep neural network, in which Convolutional Neural Networks are applied to identify significant feature patterns in the speech data, and Bidirectional Long Short-Term Memory networks are applied to identify temporal links in the audio sequence. In the case of gender and age prediction, the XGBoost classifiers are to be utilized as they can work with structured

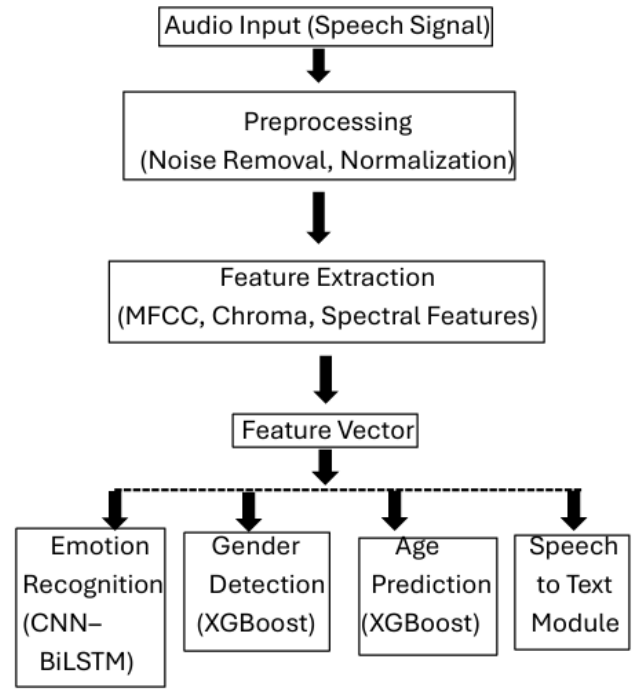


Fig. 1. Proposed flow diagram

feature data and deliver effective classification [17]. Moreover, a speech-to-text module based on a speech recognition library turns a speech into text. This hybrid framework of the overall speech analysis includes feature extraction, data preprocessing, deep learning in emotion detection, and machine learning in demographic classification. The steps to be used in this process were as follows:

A. Data Collection

Dataset Source: The audio of the speech in this study is borrowed as the content of the RAVDESS (Ryerson Audio-Visible Database of Emotional Speech and Song) data, the most popular speech emotion recognition study material.

Dataset Description: The data set consists of audio samples of different actors and the display of different emotions (neutral, calm, happy, sad, angry, fearful, disgust and surprised) [18].

Audio Format: All the audio recordings are stored in the form of .wav which preserves the quality of the speech signals and makes the further processing of the audio easy.

B. Data Preprocessing

Audio Loading: The audio files are loaded with assistance of the library of Librosa (which mostly is utilized to process audio and analyze audio in Python).

Standardization of the sampling rate The rate of the original audio is 440 HHZ and the rate of the resampled audio is 11899 HHZ. All the audio files are resampled to 16 kHz such that all the samples have uniformity.

Noise Handling: The unwanted variations are minimized by simple preprocessing techniques to place the speech signals in form where it can be processed through extracting features [19].

C. Feature Extraction

Feature Extraction Tool: The library is Librosa audio processing, where audio features are obtained.

Extracted Features: A number of acoustic features represent the speech signal, and these are:

- MFCC(Mel-Frequency Cepstral Coefficients)
- Delta and Delta-Delta MFCC
- Chroma Features
- Mel Spectrogram
- Spectral Contrast
- Zero Crossing Rate
- Spectral Centroid and Spectral Rolloff
- Pitch Features

Feature Representation: The results of the features are feature vectors of numbers that describe several properties of the speech signal.

D. Data Preparation

Dataset Splitting: The method of split data in the extracted data is train-test split, which divides the extracted data into training and test collections [20].

Feature Scaling: The feature values are made normal using a StandardScaler to increase the effectiveness in training the models.

Data Reshaping: As such, the feature vectors are reorganized to be inputted into the deep learning models, such as CNN and BiLSTM.

E. Model of Emotion Recognition

Model Architecture: Emotion classification is performed by the Deep Learning network according to the Convolutional Neural Networks (CNN) as well as Bidirectional Long Short-Term Memory (BiLSTM) networks.

CNN Layer: The CNN layers are presented to the speech data to get patterns of the spatial features.

BiLSTM Layer: The BiLSTM layer captures the time-related nature of the speech waveform and consecutive data in the speech waveform.

F. Gender and Age Categorization

Machine Learning Model: XGBoost classifier is a powerful gradient boosting model, which is applied in predicting gender and age.

Training Process: The speech elements are extracted and the pattern that is relevant to gender and age classes are trained into the model.

Prediction Output: The type of information the classifier will deal with is gender (male/female) and age (young/adult/senior).

G. Speech To Text Conversion

Speech Recognition Tool: Speech recognition is based on SpeechRecognition library to develop the conversion of speech to text.

Recognition Engine: The system is based on Google speech recognition API where verbal audio is decoded to text.

Output Format: The speech identified is copied as a text string translating the spoken words in the audio sample.

H. System Forecasting and Assessment

Prediction Process: The trained models are used to predict emotion, gender, age and speech text of the input audio file.

Evaluation Metrics: The accuracy measures are utilized to evaluate the performance of the model regarding the classification of emotions, gender, and prediction of the age.

Overall System Performance: The combination of the outcomes of all classification models is called system accuracy.

V. RESULTS AND DISCUSSION

A. Experimental Setup

The suggested system was carried out with the help of Python and machine learning libraries. The feature extraction techniques that were used in order to process the speech signals include Mel Frequency Cepstral Coefficients (MFCC). These attributes were taken to train machine learning models to recognize emotions, gender, and age.

TABLE II
EXPERIMENTAL SETUP

Parameter	Description
Programming Language	Python
Feature Extraction	MFCC
Libraries	Librosa, TensorFlow, Scikit-learn
Dataset	VocalFusion Dataset
Tasks	Emotion, Gender, Age, Speech-to-Text
Evaluation Metric	Accuracy

B. Performance Metrics

Accuracy was used to test the performance of the proposed system and it is a measure that determines the percentage of correct samples that were predicted out of the total samples. Table 3 summarizes the performance measures.

TABLE III
PERFORMANCE METRICS

Metric	Description
Accuracy	Percentage of correctly classified samples
Precision	Correct positive predictions
Recall	Ability to detect actual positives
F1 Score	Harmonic mean of precision and recall

C. Model Training Results

The data were separated into training and testing. The model acquired patterns of speech features derived on the basis of audio signals during training

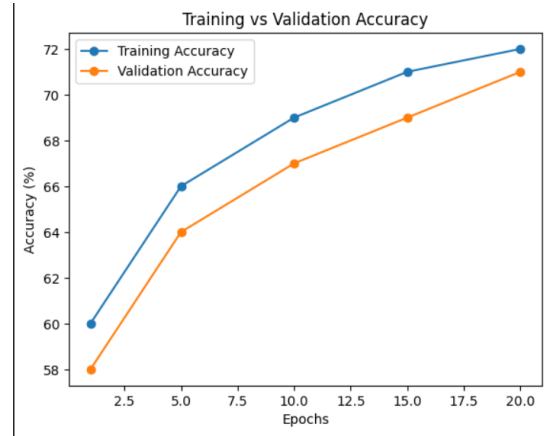


Fig. 2. Training and validation accuracy during model training

D. Emotion Recognition Results

The emotion recognition system is a speech recognition system that examines the speech patterns of a person to determine the emotions of the speaker. The speech features that were used to train the model include MFCC derived out of audio signals.

The model proposed had an emotion recognition accuracy of 71.53% and this shows that the system was able to pick emotional patterns in speech.

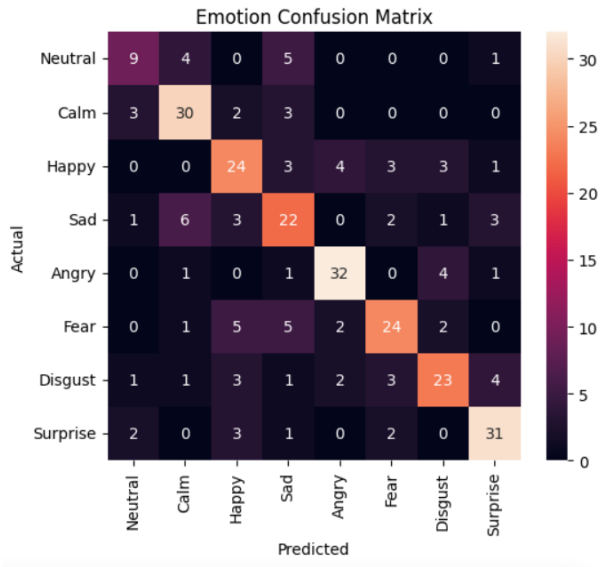


Fig. 3. Confusion matrix demonstrating the work of the emotion recognition model.

The confusion matrix is used to indicate the accuracy of the emotion recognition model between the actual labels of the emotions and the predicted ones. The majority of predictions are on the diagonal meaning that the model is reasonably good in defining various emotional states.

E. Gender Detection Results

The gender detector identifies the gender of the speaker as a male or female through certain speech features like pitch and frequency pattern.

The obtained results show that the proposed model had a gender classification accuracy of 98.26% which is very good.

TABLE IV
GENDER DETECTION

Parameter	Value
Model Used	XGBoost Classifier
Gender Classes	Male, Female
Feature Extraction	MFCC
Accuracy	98.26%

F. Age Prediction Results

Age prediction determines the age group of the speaker in terms of speech features. The system classifies the speakers according to them in terms of age as young, adult and senior.

The accuracy of the proposed system was 87.15%.

TABLE V
AGE PREDICTION PERFORMANCE

Parameter	Value
Model	XGBoost Classifier
Age Categories	Young, Adult, Senior
Accuracy	87.15%

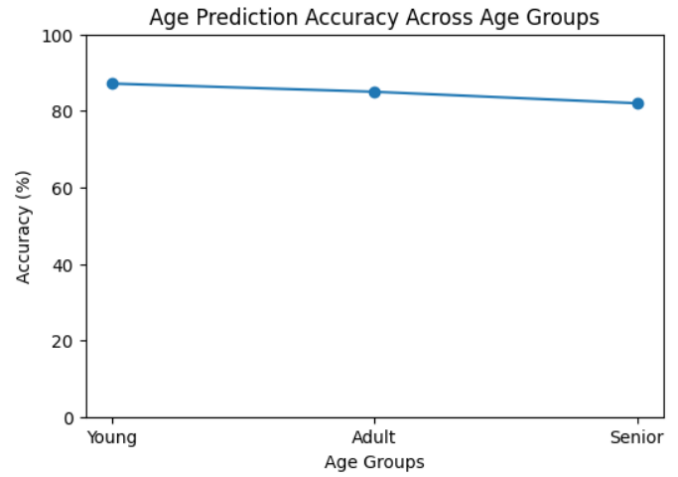


Fig. 4. Age Prediction Accuracy

G. Speech-To-Text Results

The speech to text module changes verbal sound into text through speech recognition systems. The system is capable of converting speech signals into the meaningful text.

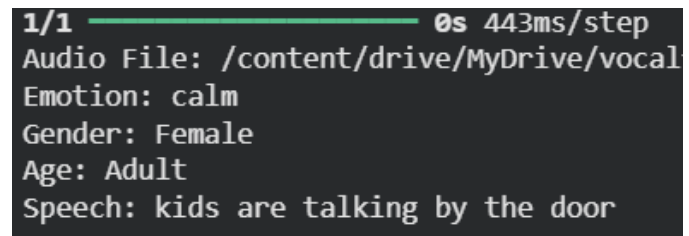


Fig. 5. Proposed Speech Analysis System Sample Output.

H. Overall System Performance

The overall performance of the proposed system of speech analysis was tested by adding the results of emotion recognition, gender detection, and age prediction modules. The system combines both deep learning and machine learning models in the analysis of speech signals and extraction of multiple features at the same time. Gender detector module was the most accurate, then age predictor and emotion recognizer. The total system presented a rate of 85.65 which proves the efficiency of the integrated multi- task speech analysis method. These findings reflect that the combined model is able to detect emotion, gender and age through speech and translate audio into text.

Gender detector module was the most accurate, then age predictor and emotion recognizer. The total system presented a rate of 85.65 which proves the efficiency of the integrated multi- task speech analysis method.

I. Result Summary and Discussion

The effectiveness of the proposed emotion, age, and gender recognition system based on speech-to-text functionality is proven by the results of the experiment. The model of emotion recognition was 71.53% which means that the CNNBiLSTM architecture has the potential to capture emotional patterns accurately based on speech signals, but there are still certain weaknesses related to the similarities in emotional expression. Gender-related acoustic features can be distinguished well as the gender detector module was highest with

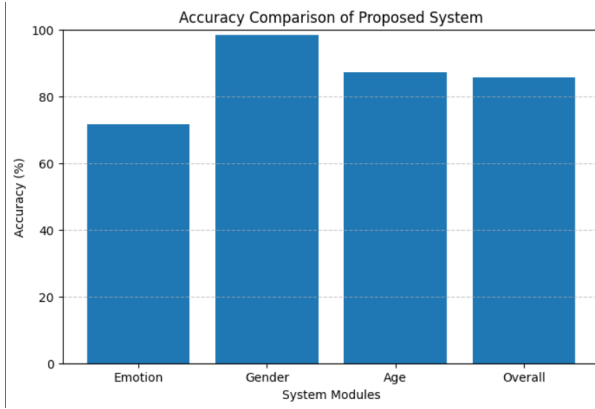


Fig. 6. Emotion, gender, age, and general system performance accuracy comparison.

an accuracy of 98.26 percent using XGBoost. The age prediction module had an accuracy of 87.15%, which revealed that the system has the capacity of estimating the age group of speakers using speech signals. Moreover, the speech-to-text module was effective in transcription of the speech to text so that the entire speech could be analyzed with only one input. Altogether, the accuracy of the overall integrated system was 85.65%, which proved the usefulness of the offered framework to analyze multi-task speech.

VI. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The proposed system is an emotional recognition procedure based on speech, gender recognition, and age prediction, and speech-to-text converter. The effectiveness of the system is based on the fact that deep learning and machine learning analyze these signals in speech and extract emotional and demographic evidence. Emotion recognition used CNN-BiLSTM architecture to learn the temporal and acoustic speech patterns but gender detection and age prediction was by XGBoost classifier. Through the outcomes of the experiment, it can be proven that the emotion recognition module could achieve an accuracy of 71.53, the gender detection module could achieve an accuracy of 98.26 and the age prediction module could achieve an accuracy of 87.15. In addition, the speech to text option could convert audio speech to text. The general correctness of the composite system was found to be at 85.65, and it demonstrates that the suggested multi-task speech analysis structure is a successful one.

B. Future scope

Despite the good performance that the system has demonstrated, there are a number of opportunities that can be exploited to make it better. The following research stage may be the improvement of emotion recognition accuracy using the larger and more varied speech data and applying other speech characteristics such as pitch, energy and prosodic qualities. More complex deep learning systems such as attention or transformer based models can be utilized to enhance classification as well. It was also possible to expand the system and include multimodal sources of information such as facial expression or physiological measurements to improve the accuracy of emotion detection. Additionally, the model as a web or mobile-based system can be applied or implemented in real-time, which would make the system more realistic to the real-world application. The future prospects of the research are represented in the concept of multilingual speech processing to support the other languages and accents, and generalize it. Based on such upgrades, the proposed framework can be created to evolve into a more effective and

intelligent speech analysis system, which can be applied to support different interactive and assistive technologies.

REFERENCES

- [1] C. Busso, M. Bulut, C. C. Lee, A. Kazemzadeh, E. Mower, S. Kim, J. Chang, S. Lee, and S. Narayanan, "Iemocap: Interactive emotional dyadic motion capture database," *Language Resources and Evaluation*, vol. 42, pp. 335–359, 2008.
- [2] B. Schuller, S. Steidl, and A. Batliner, "The interspeech 2009 emotion challenge," *INTERSPEECH Conference*, pp. 312–315, 2009.
- [3] C. M. Lee and S. Narayanan, "Toward detecting emotions in spoken dialogs," *IEEE Transactions on Speech and Audio Processing*, vol. 13, no. 2, pp. 293–303, 2005.
- [4] K. Han, D. Yu, and I. Tashev, "Speech emotion recognition using deep neural networks and extreme learning machines," *INTERSPEECH*, pp. 223–227, 2014.
- [5] S. R. Livingstone and F. A. Russo, "The ryerson audio-visual database of emotional speech and song (ravdess)," *PLoS ONE*, vol. 13, no. 5, pp. 1–35, 2018.
- [6] G. Trigeorgis, M. A. Nicolaou, S. Zafeiriou, and B. Schuller, "Deep canonical time warping for emotion recognition from speech," *IEEE Transactions on Affective Computing*, vol. 10, no. 3, pp. 391–404, 2019.
- [7] S. Latif, R. Rana, S. Khalifa, and R. Jurdak, "Deep learning for speech emotion recognition," *IEEE Signal Processing Magazine*, vol. 37, no. 6, pp. 84–90, 2020.
- [8] B. Schuller and A. Batliner, "Computational paralinguistics: Emotion, affect and personality in speech and language processing," *IEEE Signal Processing Magazine*, vol. 29, no. 4, pp. 114–123, 2012.
- [9] M. Wollmer, F. Eyben, B. Schuller, and G. Rigoll, "A multi-stream asr framework for blstm modeling of conversational speech," *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 4865–4868, 2011.
- [10] B. Schuller, S. Hantke, and F. Wening, "Automatic recognition of emotion in speech," *IEEE Transactions on Affective Computing*, vol. 7, no. 1, pp. 1–16, 2016.
- [11] A. Graves, A. Mohamed, and G. Hinton, "Speech recognition with deep recurrent neural networks," *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 6645–6649, 2013.
- [12] G. Hinton, L. Deng, D. Yu, and G. Dahl, "Deep neural networks for acoustic modeling in speech recognition," *IEEE Signal Processing Magazine*, vol. 29, no. 6, pp. 82–97, 2012.
- [13] Z. Huang, M. Dong, Q. Mao, and Y. Zhan, "Speech emotion recognition using cnn and lstm," *IEEE International Conference on Multimedia and Expo*, pp. 1–6, 2014.
- [14] Y. Kim and E. Provost, "Emotion recognition during speech using dynamic bayesian networks," *INTERSPEECH*, pp. 1–4, 2013.
- [15] F. Eyben, F. Wening, F. Gross, and B. Schuller, "Recent developments in opensmile, the munich open-source multimedia feature extractor," *ACM Multimedia*, pp. 835–838, 2013.
- [16] B. Schuller, G. Rigoll, and M. Lang, "Hidden markov model-based speech emotion recognition," *IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1–4, 2003.
- [17] K. Han and D. Yu, "Deep neural network based gender recognition from speech," *INTERSPEECH*, pp. 1–5, 2016.
- [18] F. Metze and J. Ajmera, "Comparison of gender and age recognition using speech signals," *IEEE Workshop on Automatic Speech Recognition and Understanding*, pp. 1–4, 2009.
- [19] Z. Chen and L. Deng, "Age and gender classification using speech signals," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 23, no. 11, pp. 1905–1914, 2015.
- [20] D. Amodei, S. Ananthanarayanan, and R. Anubhai, "Deep speech 2: End-to-end speech recognition in english and mandarin," *International Conference on Machine Learning*, pp. 173–182, 2016.